# Data Cleaning and Transformation

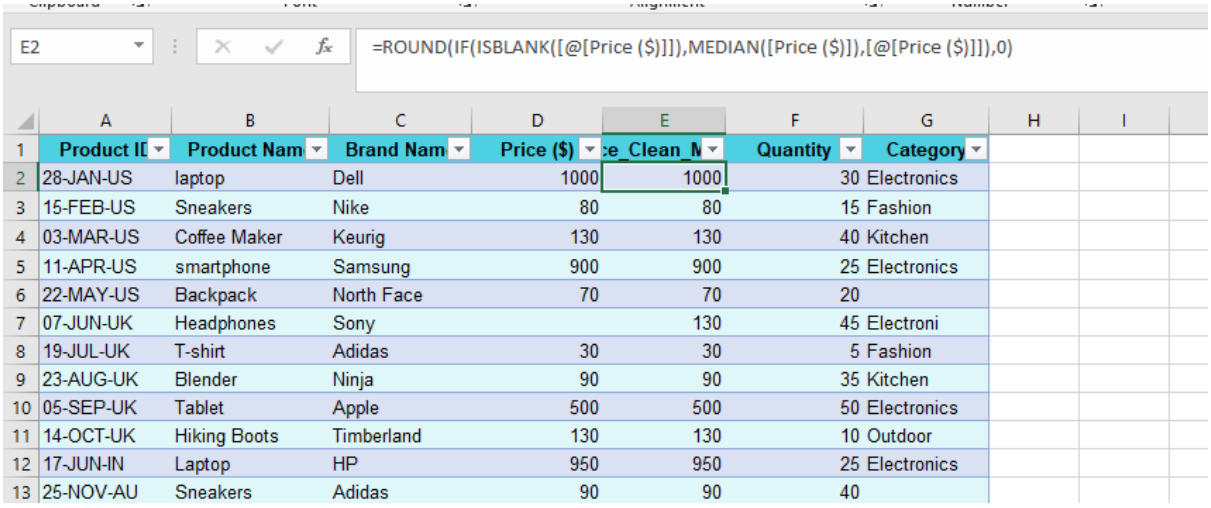
## Raw Dataset:

| Product ID | Product Name | Brand Name | Price ($) | Quantity | Category |
|---|---|---|---|---|---|
| 28-JAN-US | laptop | Dell | 1000 | 30 | Electronics |
| 15-FEB-US | Sneakers | Nike | 80 | 15 | Fashion |
| 03-MAR-US | Coffee Maker | Keurig | 130 | 40 | Kitchen |
| 11-APR-US | smartphone | Samsung | 900 | 25 | Electronics |
| 22-MAY-US | Backpack | North Face | 70 | 20 | |
| 07-JUN-UK | Headphones | Sony | | 45 | Electroni |
| 19-JUL-UK | T-shirt | Adidas | 30 | 5 | Fashion |
| 23-AUG-UK | Blender | Ninja | 90 | 35 | Kitchen |
| 05-SEP-UK | Tablet | Apple | 500 | 50 | Electronics |
| 14-OCT-UK | Hiking Boots | Timberland | 130 | 10 | Outdoor |
| 17-JUN-IN | Laptop | HP | 950 | 25 | Electronics |
| 25-NOV-AU | Sneakers | Adidas | 90 | 40 | |
| 08-DEC-DE | Coffee Maker | Nespresso | 120 | 35 | |
| 18-FEB-CA | Smartwatch | Fitbit | 150 | 15 | Electroni |
| 16-APR-ES | headphones | Bose | 250 | 20 | Electroni |
| 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories |
| 20-AUG-CN | Smartwatch | Huawei | 160 | 15 | Electroni |
| 27-JAN-IT | laptop | Asus | 980 | 10 | Electronics |
| 01-MAR-UK | Sunglasses | Oakley | 150 | 15 | Fashion |
| 14-AUG-US | Camping Tent | Coleman | | 10 | Outdoor |
| 14-MAY-RU | Camera | Nikon | 700 | 50 | Electronics |
| 09-JAN-CA | Microwave | Panasonic | 80 | 20 | Kitchen |
| 19-JUL-BR | Fitness Tracker | Xiaomi | 150 | 30 | |
| 21-AUG-CA | Laptop Bag | Samsonite | 50 | 35 | Accessories |
| 29-SEP-CA | Smartphone | Google | 800 | 45 | Electronics |
| 03-JUN-CA | Sunglasses | Ray-Ban | | 25 | Fashion |
| 11-JUL-CA | Blender | Vitamix | 400 | 40 | Kitchen |
| 16-APR-ES | headphones | Bose | 250 | 20 | Electroni |
| 07-MAR-CA | Dress | Zara | 60 | 30 | Fashion |
| 13-APR-CA | Toaster | Hamilton | 40 | 10 | Kitchen |
| 24-MAY-CA | Fitness Tracker | Garmin | 130 | 5 | Electronics |
| 02-DEC-CA | Jeans | Levi's | 50 | 50 | Fashion |
| 17-JUN-IN | Laptop | HP | 950 | 25 | Electronics |
| 09-JUL-FR | Watch | Casio | 100 | 20 | Accessories |

# Data Cleaning and Transformation

## 1. Handling Missing Values:

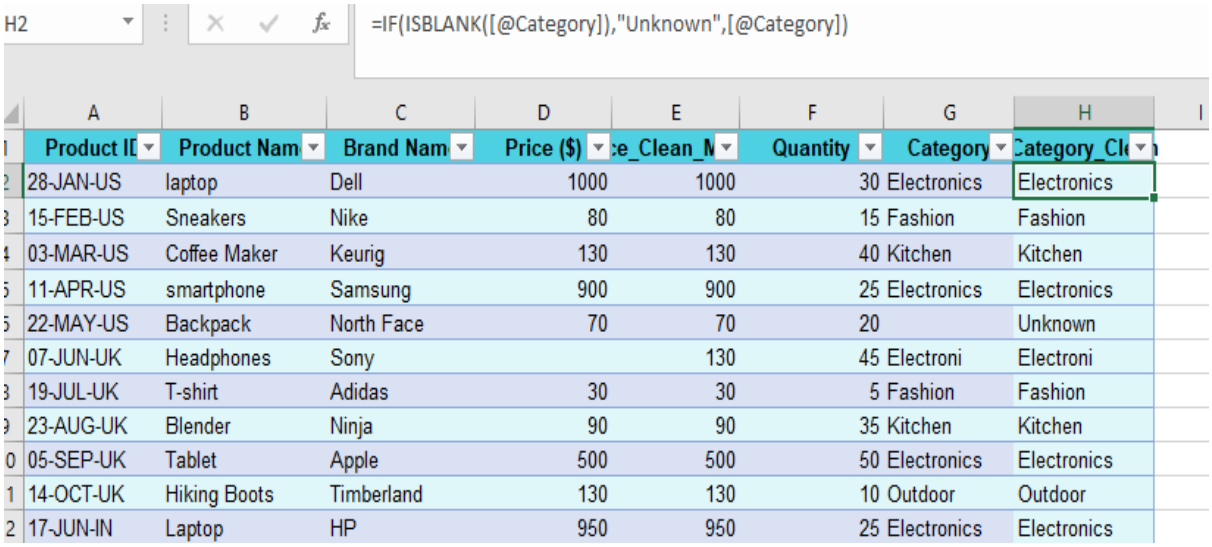
● Check for missing values in the 'Price' column. How would you handle products with missing price information?



E2 =ROUND(IF(ISBLANK([@[Price ($)]]),MEDIAN([Price ($)]),[@[Price ($)]]),0)

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Product ID | Product Nam | Brand Nam | Price ($) | e_Clean_M | Quantity | Category | | |
| 2 | 28-JAN-US | laptop | Dell | 1000 | 1000 | 30 | Electronics | | |
| 3 | 15-FEB-US | Sneakers | Nike | 80 | 80 | 15 | Fashion | | |
| 4 | 03-MAR-US | Coffee Maker | Keurig | 130 | 130 | 40 | Kitchen | | |
| 5 | 11-APR-US | smartphone | Samsung | 900 | 900 | 25 | Electronics | | |
| 6 | 22-MAY-US | Backpack | North Face | 70 | 70 | 20 | | | |
| 7 | 07-JUN-UK | Headphones | Sony | | 130 | 45 | Electroni | | |
| 8 | 19-JUL-UK | T-shirt | Adidas | 30 | 30 | 5 | Fashion | | |
| 9 | 23-AUG-UK | Blender | Ninja | 90 | 90 | 35 | Kitchen | | |
| 10 | 05-SEP-UK | Tablet | Apple | 500 | 500 | 50 | Electronics | | |
| 11 | 14-OCT-UK | Hiking Boots | Timberland | 130 | 130 | 10 | Outdoor | | |
| 12 | 17-JUN-IN | Laptop | HP | 950 | 950 | 25 | Electronics | | |
| 13 | 25-NOV-AU | Sneakers | Adidas | 90 | 90 | 40 | | | |

● If there are products with missing categories, propose a strategy to impute or deal with these missing values effectively.



H2 =IF(ISBLANK([@Category]),"Unknown",[@Category])

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Product ID | Product Nam | Brand Nam | Price ($) | e_Clean_M | Quantity | Category | Category_Cl | |
| 2 | 28-JAN-US | laptop | Dell | 1000 | 1000 | 30 | Electronics | Electronics | |
| 3 | 15-FEB-US | Sneakers | Nike | 80 | 80 | 15 | Fashion | Fashion | |
| 4 | 03-MAR-US | Coffee Maker | Keurig | 130 | 130 | 40 | Kitchen | Kitchen | |
| 5 | 11-APR-US | smartphone | Samsung | 900 | 900 | 25 | Electronics | Electronics | |
| 6 | 22-MAY-US | Backpack | North Face | 70 | 70 | 20 | | Unknown | |
| 7 | 07-JUN-UK | Headphones | Sony | | 130 | 45 | Electroni | Electroni | |
| 8 | 19-JUL-UK | T-shirt | Adidas | 30 | 30 | 5 | Fashion | Fashion | |
| 9 | 23-AUG-UK | Blender | Ninja | 90 | 90 | 35 | Kitchen | Kitchen | |
| 10 | 05-SEP-UK | Tablet | Apple | 500 | 500 | 50 | Electronics | Electronics | |
| 11 | 14-OCT-UK | Hiking Boots | Timberland | 130 | 130 | 10 | Outdoor | Outdoor | |
| 12 | 17-JUN-IN | Laptop | HP | 950 | 950 | 25 | Electronics | Electronics | |

## 2. Correcting Inconsistent Data:

● Identify any inconsistent text formats present in the "Product Name" column.

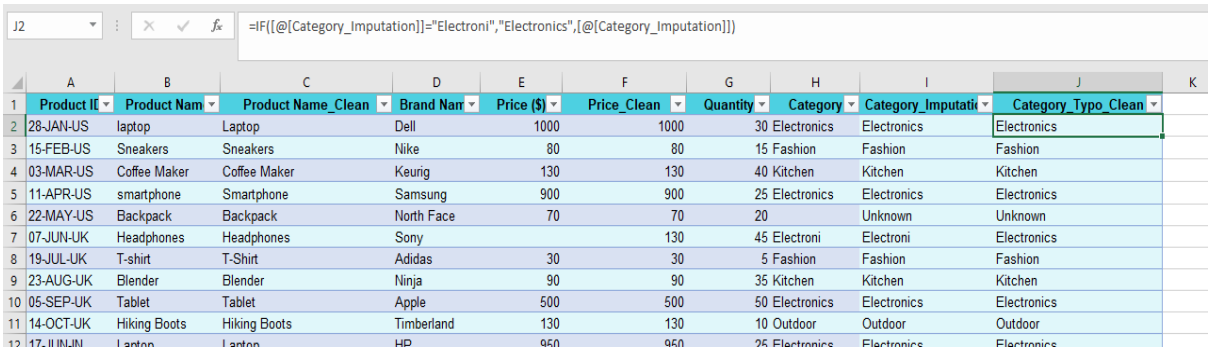


C2 =PROPER([@[Product Name]])

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Product ID | Product Nam | Product Name_Clean | Brand Nam | Price ($) | Price_Clean | Quantity | Category | Category_Clean |
| 2 | 28-JAN-US | laptop | Laptop | Dell | 1000 | 1000 | 30 | Electronics | Electronics |
| 3 | 15-FEB-US | Sneakers | Sneakers | Nike | 80 | 80 | 15 | Fashion | Fashion |
| 4 | 03-MAR-US | Coffee Maker | Coffee Maker | Keurig | 130 | 130 | 40 | Kitchen | Kitchen |
| 5 | 11-APR-US | smartphone | Smartphone | Samsung | 900 | 900 | 25 | Electronics | Electronics |
| 6 | 22-MAY-US | Backpack | Backpack | North Face | 70 | 70 | 20 | | Unknown |
| 7 | 07-JUN-UK | Headphones | Headphones | Sony | | 130 | 45 | Electroni | Electroni |
| 8 | 19-JUL-UK | T-shirt | T-Shirt | Adidas | 30 | 30 | 5 | Fashion | Fashion |
| 9 | 23-AUG-UK | Blender | Blender | Ninja | 90 | 90 | 35 | Kitchen | Kitchen |
| 10 | 05-SEP-UK | Tablet | Tablet | Apple | 500 | 500 | 50 | Electronics | Electronics |
| 11 | 14-OCT-UK | Hiking Boots | Hiking Boots | Timberland | 130 | 130 | 10 | Outdoor | Outdoor |
| 12 | 17-JUN-IN | Laptop | Laptop | HP | 950 | 950 | 25 | Electronics | Electronics |

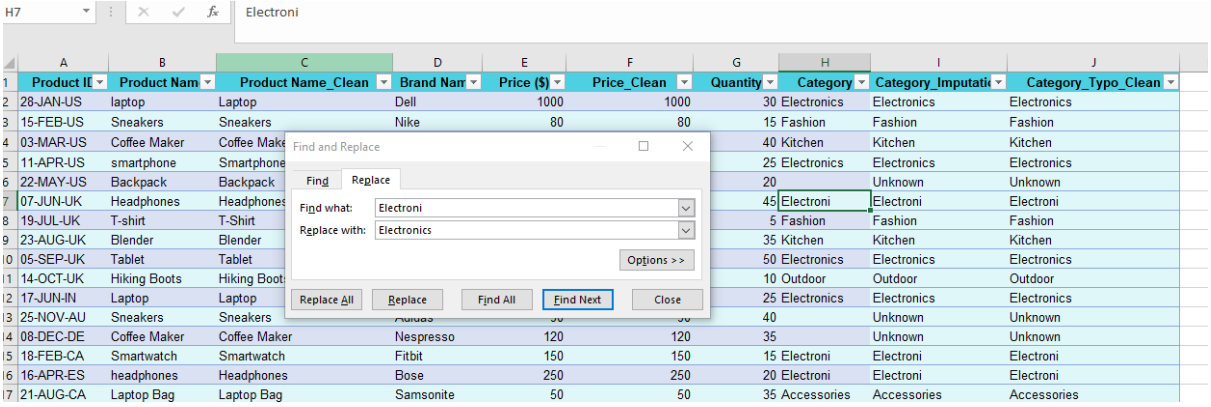
● Identify any typos present in the "Category" column.

● Use the find and replace function to standardize the text formats in the "Product Name" column and fix any typos or misspellings in the "Category" column.



J2 =IF([@[Category_Imputation]]="Electroni","Electronics",[@[Category_Imputation]])

| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Product ID | Product Nam | Product Name_Clean | Brand Nam | Price ($) | Price_Clean | Quantity | Category | Category_Imputati | Category_Typo_Clean |
| 2 | 28-JAN-US | laptop | Laptop | Dell | 1000 | 1000 | 30 | Electronics | Electronics | Electronics |
| 3 | 15-FEB-US | Sneakers | Sneakers | Nike | 80 | 80 | 15 | Fashion | Fashion | Fashion |
| 4 | 03-MAR-US | Coffee Maker | Coffee Maker | Keurig | 130 | 130 | 40 | Kitchen | Kitchen | Kitchen |
| 5 | 11-APR-US | smartphone | Smartphone | Samsung | 900 | 900 | 25 | Electronics | Electronics | Electronics |
| 6 | 22-MAY-US | Backpack | Backpack | North Face | 70 | 70 | 20 | | Unknown | Unknown |
| 7 | 07-JUN-UK | Headphones | Headphones | Sony | | 130 | 45 | Electroni | Electroni | Electronics |
| 8 | 19-JUL-UK | T-shirt | T-Shirt | Adidas | 30 | 30 | 5 | Fashion | Fashion | Fashion |
| 9 | 23-AUG-UK | Blender | Blender | Ninja | 90 | 90 | 35 | Kitchen | Kitchen | Kitchen |
| 10 | 05-SEP-UK | Tablet | Tablet | Apple | 500 | 500 | 50 | Electronics | Electronics | Electronics |
| 11 | 14-OCT-UK | Hiking Boots | Hiking Boots | Timberland | 130 | 130 | 10 | Outdoor | Outdoor | Outdoor |
| 12 | 17-JUN-IN | Laptop | Laptop | HP | 950 | 950 | 25 | Electronics | Electronics | Electronics |



H7 Electroni

Find and Replace

Find | Replace

Find what: Electroni

Replace with: Electronics

Options >>

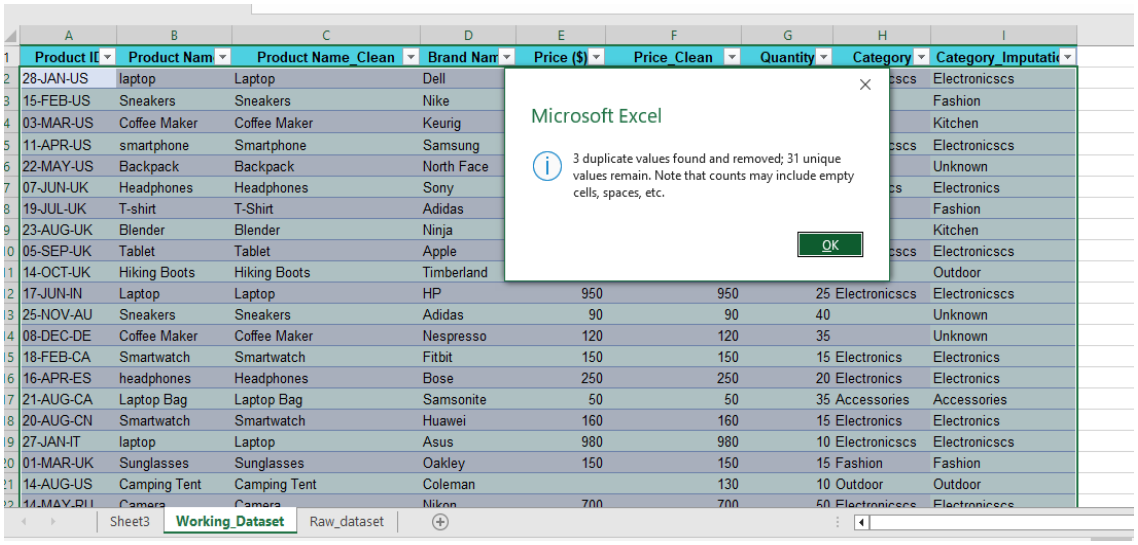
Replace All | Replace | Find All | Find Next | Close

| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Product ID | Product Nam | Product Name_Clean | Brand Nam | Price ($) | Price_Clean | Quantity | Category | Category_Imputati | Category_Typo_Clean |
| 2 | 28-JAN-US | laptop | Laptop | Dell | 1000 | 1000 | 30 | Electronics | Electronics | Electronics |
| 3 | 15-FEB-US | Sneakers | Sneakers | Nike | 80 | 80 | 15 | Fashion | Fashion | Fashion |
| 4 | 03-MAR-US | Coffee Maker | Coffee Make | | | | 40 | Kitchen | Kitchen | Kitchen |
| 5 | 11-APR-US | smartphone | Smartphone | | | | 25 | Electronics | Electronics | Electronics |
| 6 | 22-MAY-US | Backpack | Backpack | | | | 20 | | Unknown | Unknown |
| 7 | 07-JUN-UK | Headphones | Headphones | | | | 45 | Electroni | Electroni | Electroni |
| 8 | 19-JUL-UK | T-shirt | T-Shirt | | | | 5 | Fashion | Fashion | Fashion |
| 9 | 23-AUG-UK | Blender | Blender | | | | 35 | Kitchen | Kitchen | Kitchen |
| 10 | 05-SEP-UK | Tablet | Tablet | | | | 50 | Electronics | Electronics | Electronics |
| 11 | 14-OCT-UK | Hiking Boots | Hiking Boot | | | | 10 | Outdoor | Outdoor | Outdoor |
| 12 | 17-JUN-IN | Laptop | Laptop | | | | 25 | Electronics | Electronics | Electronics |
| 13 | 25-NOV-AU | Sneakers | Sneakers | | | | 40 | | Unknown | Unknown |
| 14 | 08-DEC-DE | Coffee Maker | Coffee Maker | Nespresso | 120 | 120 | 35 | | Unknown | Unknown |
| 15 | 18-FEB-CA | Smartwatch | Smartwatch | Fitbit | 150 | 150 | 15 | Electroni | Electroni | Electroni |
| 16 | 16-APR-ES | headphones | Headphones | Bose | 250 | 250 | 20 | Electroni | Electroni | Electroni |
| 17 | 21-AUG-CA | Laptop Bag | Laptop Bag | Samsonite | 50 | 50 | 35 | Accessories | Accessories | Accessories |

# Data Cleaning and Transformation

## 3. Removing Duplicates:

● Identify any duplicate rows within the dataset based on the entirety of each row, and remove them if any.

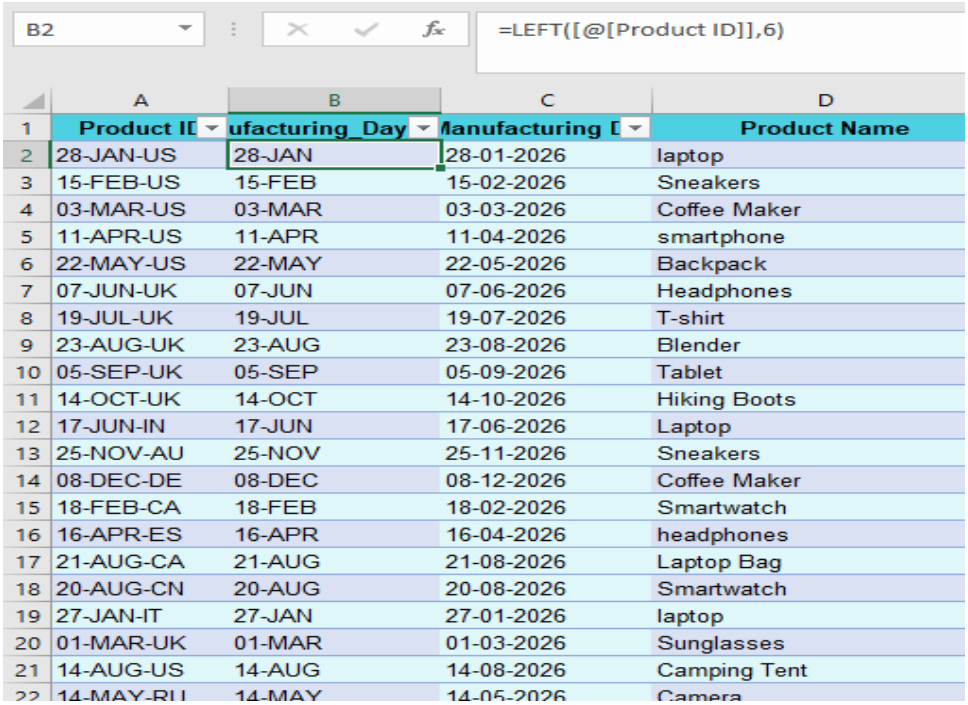



| Product ID | Product Nam | Product Name_Clean | Brand Nam | Price ($) | Price_Clean | Quantity | Category | Category_Imputati |
|---|---|---|---|---|---|---|---|---|
| 28-JAN-US | laptop | Laptop | Dell | | | | cscs | Electronicscs |
| 15-FEB-US | Sneakers | Sneakers | Nike | | | | | Fashion |
| 03-MAR-US | Coffee Maker | Coffee Maker | Keurig | | | | | Kitchen |
| 11-APR-US | smartphone | Smartphone | Samsung | | | | cscs | Electronicscs |
| 22-MAY-US | Backpack | Backpack | North Face | | | | | Unknown |
| 07-JUN-UK | Headphones | Headphones | Sony | | | | cs | Electronics |
| 19-JUL-UK | T-shirt | T-Shirt | Adidas | | | | | Fashion |
| 23-AUG-UK | Blender | Blender | Ninja | | | | | Kitchen |
| 05-SEP-UK | Tablet | Tablet | Apple | | | | cscs | Electronicscs |
| 14-OCT-UK | Hiking Boots | Hiking Boots | Timberland | | | | | Outdoor |
| 17-JUN-IN | Laptop | Laptop | HP | 950 | 950 | 25 | Electronicscs | Electronicscs |
| 25-NOV-AU | Sneakers | Sneakers | Adidas | 90 | 90 | 40 | | Unknown |
| 08-DEC-DE | Coffee Maker | Coffee Maker | Nespresso | 120 | 120 | 35 | | Unknown |
| 18-FEB-CA | Smartwatch | Smartwatch | Fitbit | 150 | 150 | 15 | Electronics | Electronics |
| 16-APR-ES | headphones | Headphones | Bose | 250 | 250 | 20 | Electronics | Electronics |
| 21-AUG-CA | Laptop Bag | Laptop Bag | Samsonite | 50 | 50 | 35 | Accessories | Accessories |
| 20-AUG-CN | Smartwatch | Smartwatch | Huawei | 160 | 160 | 15 | Electronics | Electronics |
| 27-JAN-IT | laptop | Laptop | Asus | 980 | 980 | 10 | Electronicscs | Electronicscs |
| 01-MAR-UK | Sunglasses | Sunglasses | Oakley | 150 | 150 | 15 | Fashion | Fashion |
| 14-AUG-US | Camping Tent | Camping Tent | Coleman | | 130 | 10 | Outdoor | Outdoor |
| 14-MAY-RU | Camera | Camera | Nikon | 700 | 700 | 50 | Electronicscs | Electronicscs |

## 4. Splitting and Merging Data:

● Split the "Product ID" column into two separate columns for "Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

***Type 1:***



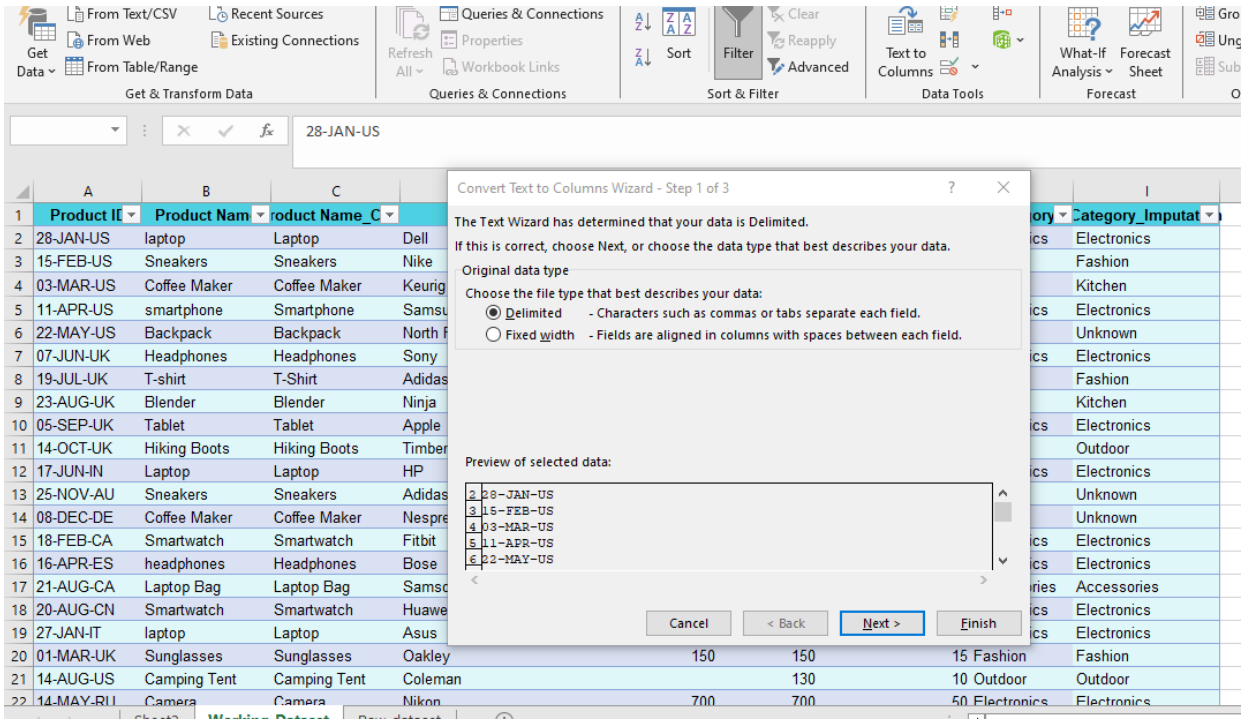

| | A | B | C | D |
|---|---|---|---|---|
| 1 | Product ID | ufacturing_Day | Manufacturing D | Product Name |
| 2 | 28-JAN-US | 28-JAN | 28-01-2026 | laptop |
| 3 | 15-FEB-US | 15-FEB | 15-02-2026 | Sneakers |
| 4 | 03-MAR-US | 03-MAR | 03-03-2026 | Coffee Maker |
| 5 | 11-APR-US | 11-APR | 11-04-2026 | smartphone |
| 6 | 22-MAY-US | 22-MAY | 22-05-2026 | Backpack |
| 7 | 07-JUN-UK | 07-JUN | 07-06-2026 | Headphones |
| 8 | 19-JUL-UK | 19-JUL | 19-07-2026 | T-shirt |
| 9 | 23-AUG-UK | 23-AUG | 23-08-2026 | Blender |
| 10 | 05-SEP-UK | 05-SEP | 05-09-2026 | Tablet |
| 11 | 14-OCT-UK | 14-OCT | 14-10-2026 | Hiking Boots |
| 12 | 17-JUN-IN | 17-JUN | 17-06-2026 | Laptop |
| 13 | 25-NOV-AU | 25-NOV | 25-11-2026 | Sneakers |
| 14 | 08-DEC-DE | 08-DEC | 08-12-2026 | Coffee Maker |
| 15 | 18-FEB-CA | 18-FEB | 18-02-2026 | Smartwatch |
| 16 | 16-APR-ES | 16-APR | 16-04-2026 | headphones |
| 17 | 21-AUG-CA | 21-AUG | 21-08-2026 | Laptop Bag |
| 18 | 20-AUG-CN | 20-AUG | 20-08-2026 | Smartwatch |
| 19 | 27-JAN-IT | 27-JAN | 27-01-2026 | laptop |
| 20 | 01-MAR-UK | 01-MAR | 01-03-2026 | Sunglasses |
| 21 | 14-AUG-US | 14-AUG | 14-08-2026 | Camping Tent |
| 22 | 14-MAY-RU | 14-MAY | 14-05-2026 | Camera |

# Data Cleaning and Transformation

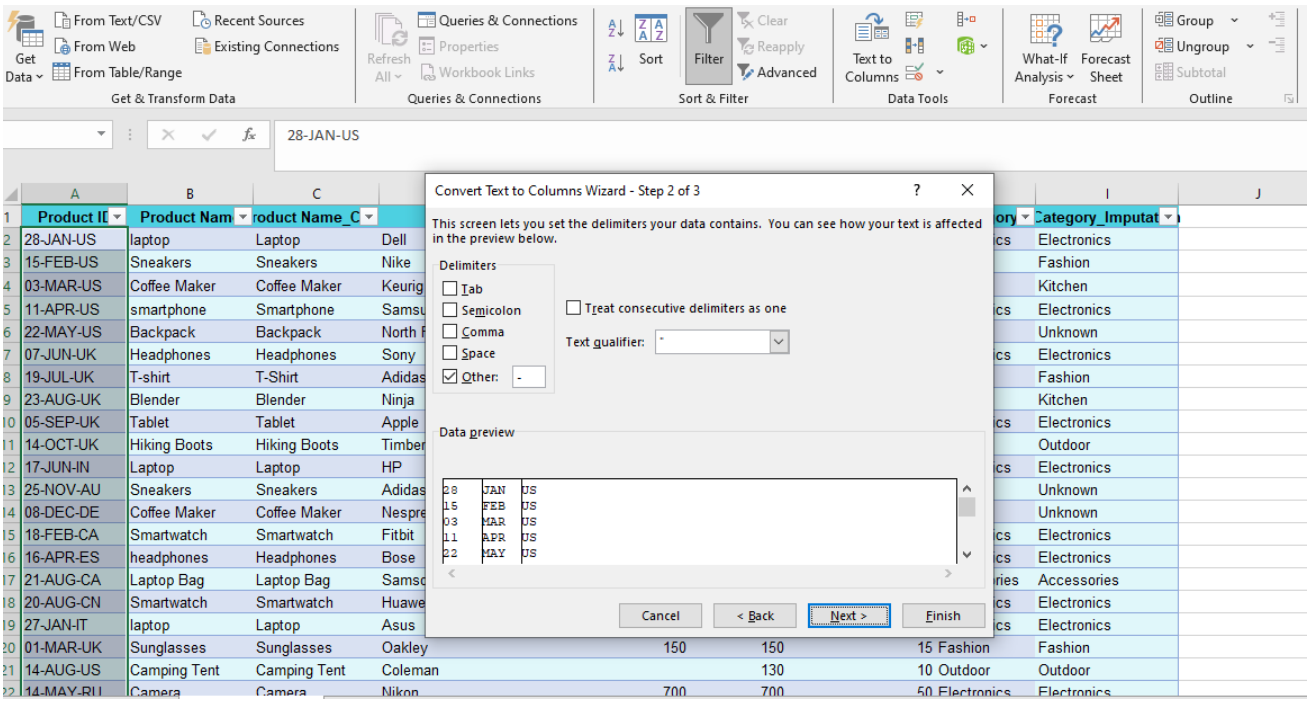
C2 =RIGHT([@[Product ID]],2)

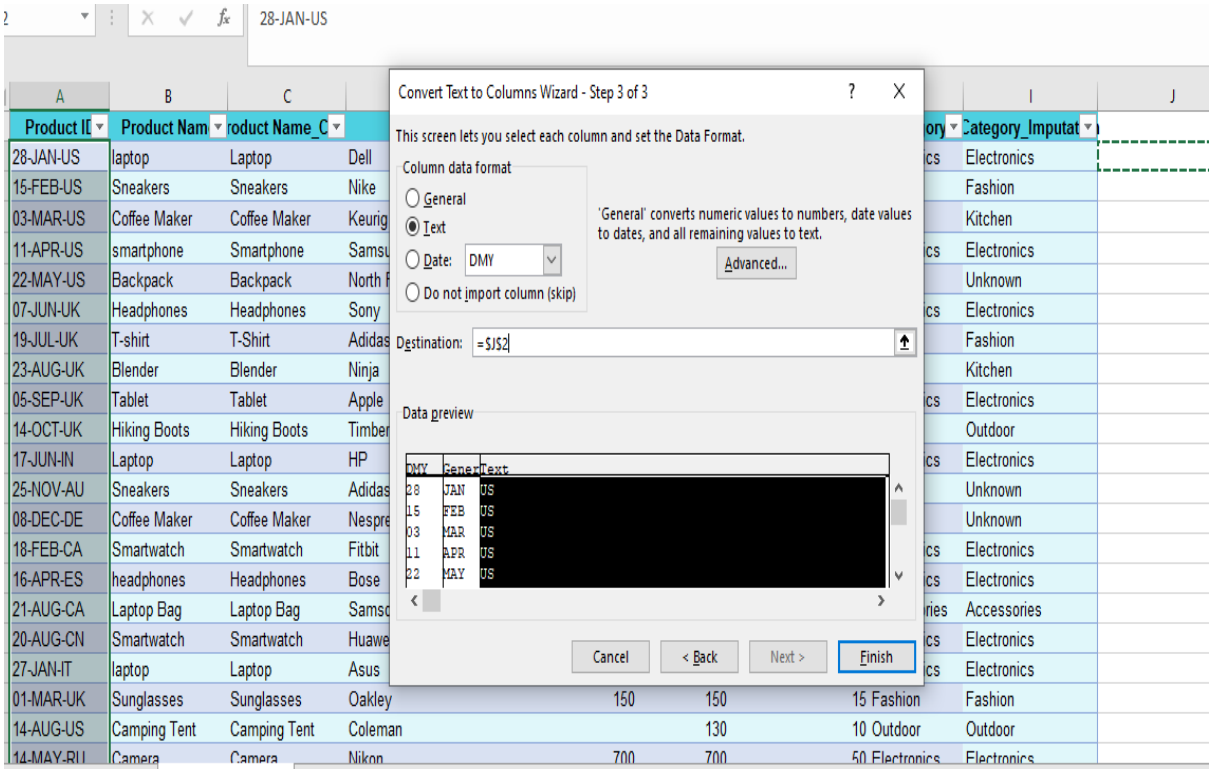
| | A | B | C | D |
|---|---|---|---|---|
| | Product ID | ufacturing_Day | Country Code | Manufacturing Date |
| 2 | 28-JAN-US | 28-JAN | US | 28-01-2026 |
| 3 | 15-FEB-US | 15-FEB | US | 15-02-2026 |
| 4 | 03-MAR-US | 03-MAR | US | 03-03-2026 |
| 5 | 11-APR-US | 11-APR | US | 11-04-2026 |
| 6 | 22-MAY-US | 22-MAY | US | 22-05-2026 |
| 7 | 07-JUN-UK | 07-JUN | UK | 07-06-2026 |
| 8 | 19-JUL-UK | 19-JUL | UK | 19-07-2026 |
| 9 | 23-AUG-UK | 23-AUG | UK | 23-08-2026 |
| 10 | 05-SEP-UK | 05-SEP | UK | 05-09-2026 |
| 11 | 14-OCT-UK | 14-OCT | UK | 14-10-2026 |
| 12 | 17-JUN-IN | 17-JUN | IN | 17-06-2026 |
| 13 | 25-NOV-AU | 25-NOV | AU | 25-11-2026 |
| 14 | 08-DEC-DE | 08-DEC | DE | 08-12-2026 |
| 15 | 18-FEB-CA | 18-FEB | CA | 18-02-2026 |
| 16 | 16-APR-ES | 16-APR | ES | 16-04-2026 |
| 17 | 21-AUG-CA | 21-AUG | CA | 21-08-2026 |
| 18 | 20-AUG-CN | 20-AUG | CN | 20-08-2026 |
| 19 | 27-JAN-IT | 27-JAN | IT | 27-01-2026 |
| 20 | 01-MAR-UK | 01-MAR | UK | 01-03-2026 |
| 21 | 14-AUG-US | 14-AUG | US | 14-08-2026 |
| 22 | 14-MAY-RU | 14-MAY | RU | 14-05-2026 |

***Type 2:***



# Data Cleaning and Transformation

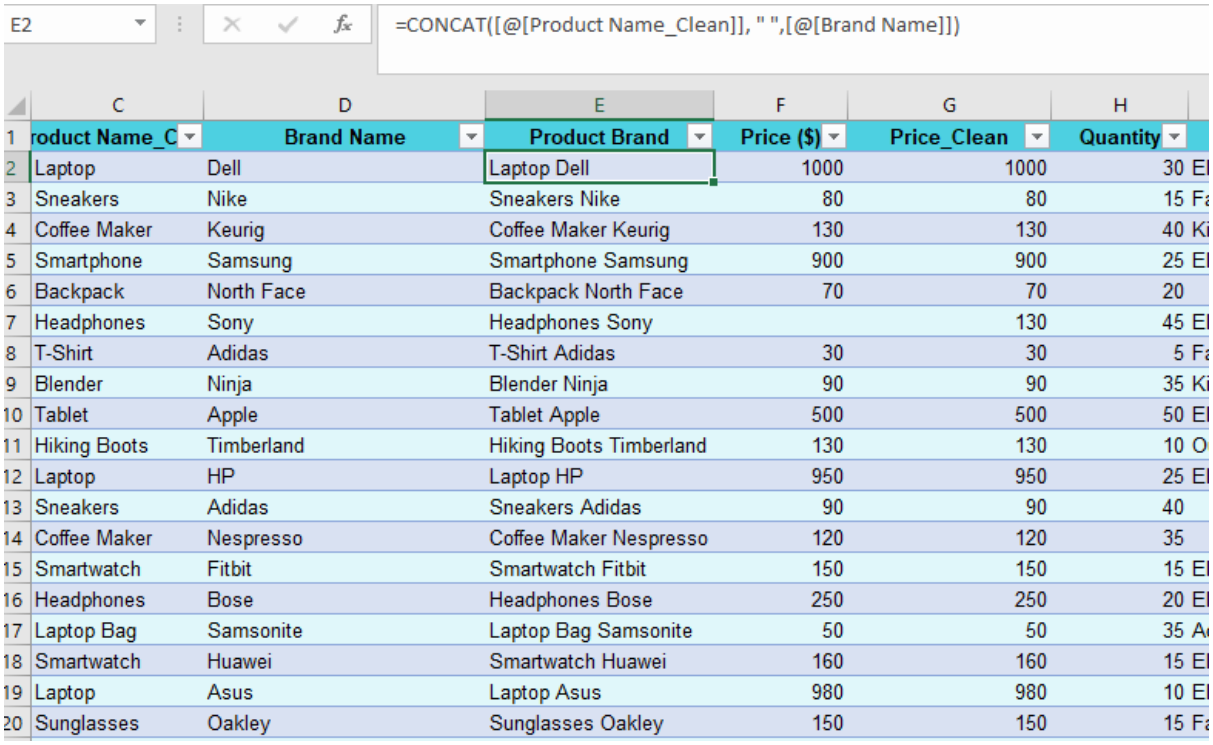


# Data Cleaning and Transformation

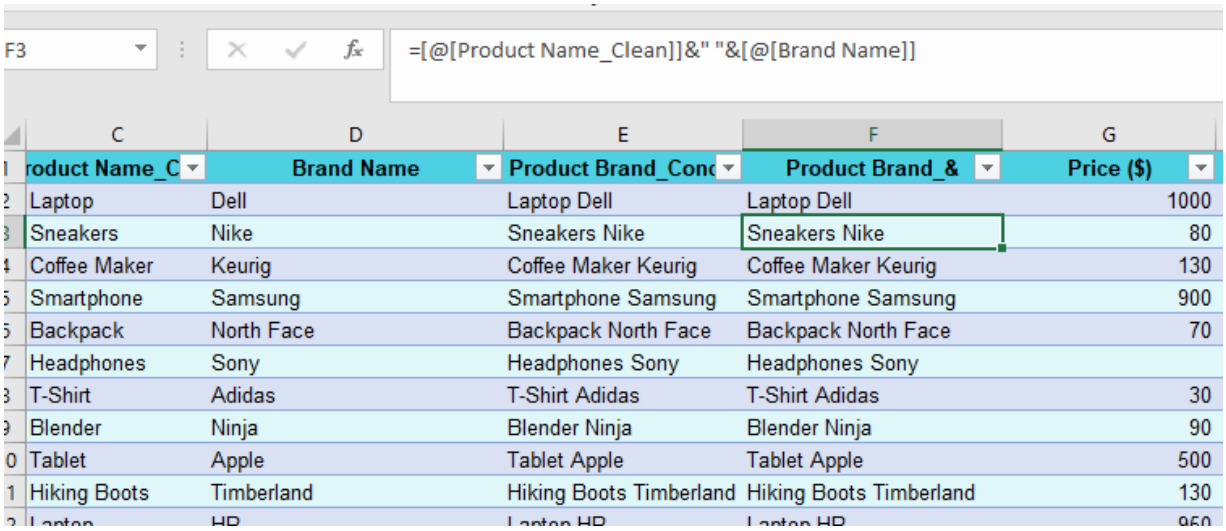
● Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

***Type 1:***




| | C | D | E | F | G | H |
|---|---|---|---|---|---|---|
| 1 | roduct Name_C | Brand Name | Product Brand | Price ($) | Price_Clean | Quantity |
| 2 | Laptop | Dell | Laptop Dell | 1000 | 1000 | 30 El |
| 3 | Sneakers | Nike | Sneakers Nike | 80 | 80 | 15 Fa |
| 4 | Coffee Maker | Keurig | Coffee Maker Keurig | 130 | 130 | 40 Ki |
| 5 | Smartphone | Samsung | Smartphone Samsung | 900 | 900 | 25 El |
| 6 | Backpack | North Face | Backpack North Face | 70 | 70 | 20 |
| 7 | Headphones | Sony | Headphones Sony | | 130 | 45 El |
| 8 | T-Shirt | Adidas | T-Shirt Adidas | 30 | 30 | 5 Fa |
| 9 | Blender | Ninja | Blender Ninja | 90 | 90 | 35 Ki |
| 10 | Tablet | Apple | Tablet Apple | 500 | 500 | 50 El |
| 11 | Hiking Boots | Timberland | Hiking Boots Timberland | 130 | 130 | 10 O |
| 12 | Laptop | HP | Laptop HP | 950 | 950 | 25 El |
| 13 | Sneakers | Adidas | Sneakers Adidas | 90 | 90 | 40 |
| 14 | Coffee Maker | Nespresso | Coffee Maker Nespresso | 120 | 120 | 35 |
| 15 | Smartwatch | Fitbit | Smartwatch Fitbit | 150 | 150 | 15 El |
| 16 | Headphones | Bose | Headphones Bose | 250 | 250 | 20 El |
| 17 | Laptop Bag | Samsonite | Laptop Bag Samsonite | 50 | 50 | 35 Ac |
| 18 | Smartwatch | Huawei | Smartwatch Huawei | 160 | 160 | 15 El |
| 19 | Laptop | Asus | Laptop Asus | 980 | 980 | 10 El |
| 20 | Sunglasses | Oakley | Sunglasses Oakley | 150 | 150 | 15 Fa |

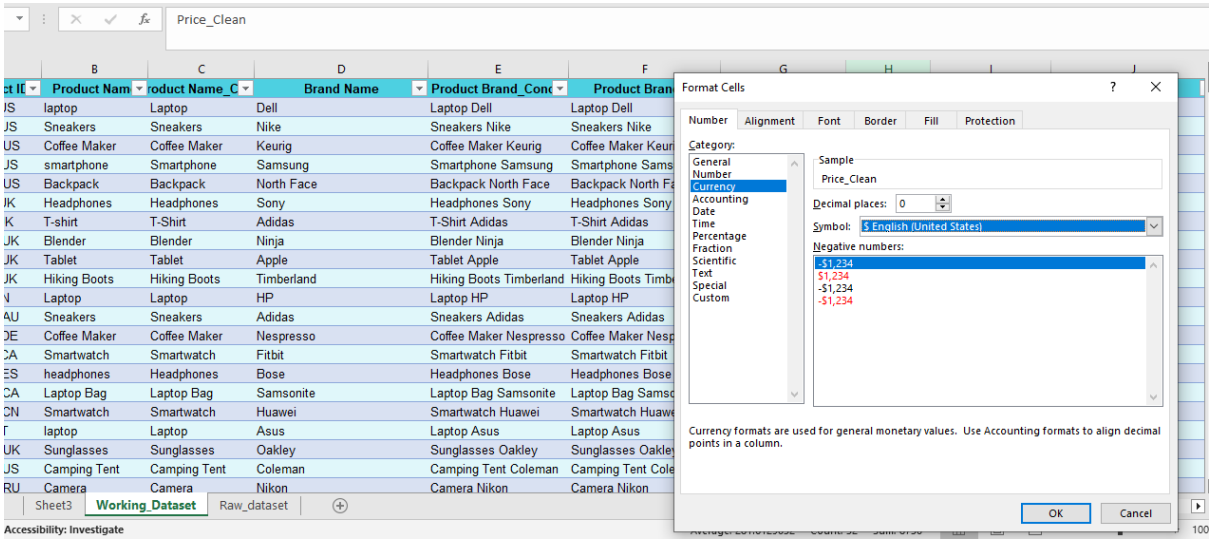
***Type 2:***




| | C | D | E | F | G |
|---|---|---|---|---|---|
| 1 | roduct Name_C | Brand Name | Product Brand_Conc | Product Brand_& | Price ($) |
| 2 | Laptop | Dell | Laptop Dell | Laptop Dell | 1000 |
| 3 | Sneakers | Nike | Sneakers Nike | Sneakers Nike | 80 |
| 4 | Coffee Maker | Keurig | Coffee Maker Keurig | Coffee Maker Keurig | 130 |
| 5 | Smartphone | Samsung | Smartphone Samsung | Smartphone Samsung | 900 |
| 6 | Backpack | North Face | Backpack North Face | Backpack North Face | 70 |
| 7 | Headphones | Sony | Headphones Sony | Headphones Sony | |
| 8 | T-Shirt | Adidas | T-Shirt Adidas | T-Shirt Adidas | 30 |
| 9 | Blender | Ninja | Blender Ninja | Blender Ninja | 90 |
| 10 | Tablet | Apple | Tablet Apple | Tablet Apple | 500 |
| 11 | Hiking Boots | Timberland | Hiking Boots Timberland | Hiking Boots Timberland | 130 |
| 12 | Laptop | HP | Laptop HP | Laptop HP | 950 |

## 5. Number Formatting:

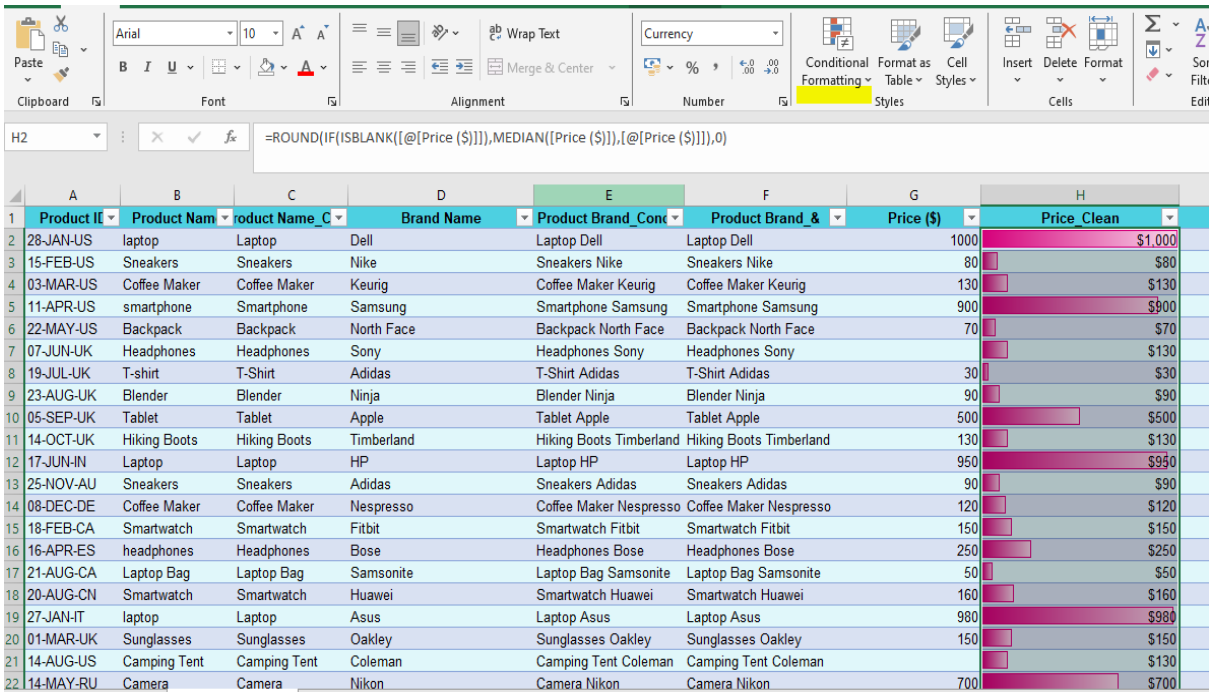
- Format the data type of the "Price" column to a currency format.




| F | G | H | I |
|---|---|---|---|
| roduct Brand_& | Price ($) | Price_Clean | Quantity |
| p Dell | 1000 | $1,000 | |
| kers Nike | 80 | $80 | |
| e Maker Keurig | 130 | $130 | |
| tphone Samsung | 900 | $900 | |
| pack North Face | 70 | $70 | |
| phones Sony | | $130 | |
| t Adidas | 30 | $30 | |
| er Ninja | 90 | $90 | |
| t Apple | 500 | $500 | |
| g Boots Timberland | 130 | $130 | |
| p HP | 950 | $950 | |
| kers Adidas | 90 | $90 | |
| e Maker Nespresso | 120 | $120 | |
| twatch Fitbit | 150 | $150 | |
| phones Bose | 250 | $250 | |
| p Bag Samsonite | 50 | $50 | |
| twatch Huawei | 160 | $160 | |
| p Asus | 980 | $980 | |
| asses Oakley | 150 | $150 | |
| ing Tent Coleman | | $130 | |
| ra Nikon | 700 | $700 | |

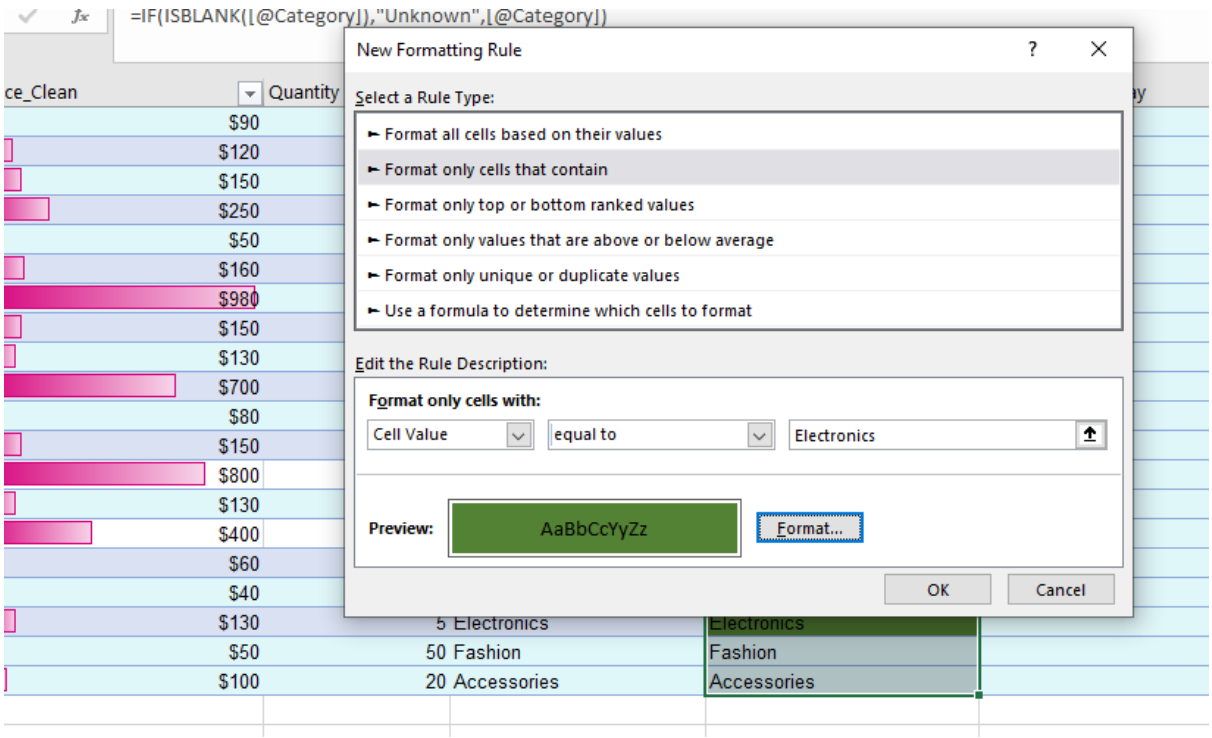
## 6. Conditional Formatting:

● Apply data bar or color scales conditional formatting in the "Price" column.

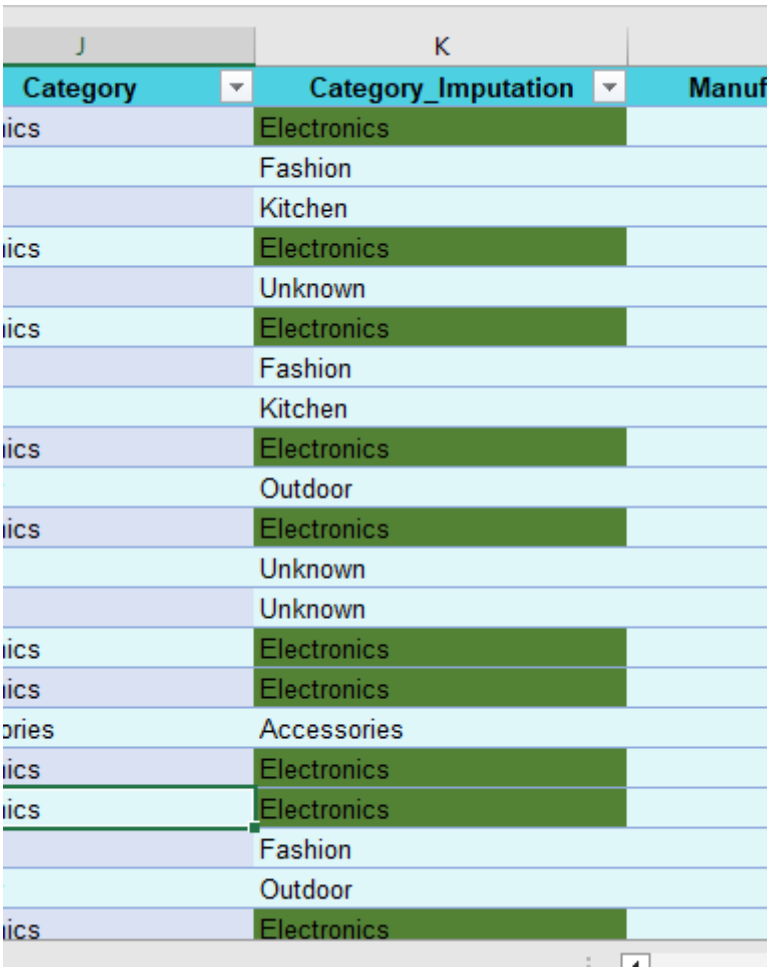



| | A | B | C | D | E | F | G | H |
|---|---|---|---|---|---|---|---|---|
| 1 | Product ID | Product Nam | roduct Name_C | Brand Name | Product Brand_Conc | Product Brand_& | Price ($) | Price_Clean |
| 2 | 28-JAN-US | laptop | Laptop | Dell | Laptop Dell | Laptop Dell | 1000 | $1,000 |
| 3 | 15-FEB-US | Sneakers | Sneakers | Nike | Sneakers Nike | Sneakers Nike | 80 | $80 |
| 4 | 03-MAR-US | Coffee Maker | Coffee Maker | Keurig | Coffee Maker Keurig | Coffee Maker Keurig | 130 | $130 |
| 5 | 11-APR-US | smartphone | Smartphone | Samsung | Smartphone Samsung | Smartphone Samsung | 900 | $900 |
| 6 | 22-MAY-US | Backpack | Backpack | North Face | Backpack North Face | Backpack North Face | 70 | $70 |
| 7 | 07-JUN-UK | Headphones | Headphones | Sony | Headphones Sony | Headphones Sony | | $130 |
| 8 | 19-JUL-UK | T-shirt | T-Shirt | Adidas | T-Shirt Adidas | T-Shirt Adidas | 30 | $30 |
| 9 | 23-AUG-UK | Blender | Blender | Ninja | Blender Ninja | Blender Ninja | 90 | $90 |
| 10 | 05-SEP-UK | Tablet | Tablet | Apple | Tablet Apple | Tablet Apple | 500 | $500 |
| 11 | 14-OCT-UK | Hiking Boots | Hiking Boots | Timberland | Hiking Boots Timberland | Hiking Boots Timberland | 130 | $130 |
| 12 | 17-JUN-IN | Laptop | Laptop | HP | Laptop HP | Laptop HP | 950 | $950 |
| 13 | 25-NOV-AU | Sneakers | Sneakers | Adidas | Sneakers Adidas | Sneakers Adidas | 90 | $90 |
| 14 | 08-DEC-DE | Coffee Maker | Coffee Maker | Nespresso | Coffee Maker Nespresso | Coffee Maker Nespresso | 120 | $120 |
| 15 | 18-FEB-CA | Smartwatch | Smartwatch | Fitbit | Smartwatch Fitbit | Smartwatch Fitbit | 150 | $150 |
| 16 | 16-APR-ES | headphones | Headphones | Bose | Headphones Bose | Headphones Bose | 250 | $250 |
| 17 | 21-AUG-CA | Laptop Bag | Laptop Bag | Samsonite | Laptop Bag Samsonite | Laptop Bag Samsonite | 50 | $50 |
| 18 | 20-AUG-CN | Smartwatch | Smartwatch | Huawei | Smartwatch Huawei | Smartwatch Huawei | 160 | $160 |
| 19 | 27-JAN-IT | laptop | Laptop | Asus | Laptop Asus | Laptop Asus | 980 | $980 |
| 20 | 01-MAR-UK | Sunglasses | Sunglasses | Oakley | Sunglasses Oakley | Sunglasses Oakley | 150 | $150 |
| 21 | 14-AUG-US | Camping Tent | Camping Tent | Coleman | Camping Tent Coleman | Camping Tent Coleman | | $130 |
| 22 | 14-MAY-RU | Camera | Camera | Nikon | Camera Nikon | Camera Nikon | 700 | $700 |

● Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."



# Data Cleaning and Transformation



| J | K | |
|---|---|---|
| Category | Category_Imputation | Manuf |
| ics | Electronics | |
| | Fashion | |
| | Kitchen | |
| ics | Electronics | |
| | Unknown | |
| ics | Electronics | |
| | Fashion | |
| | Kitchen | |
| ics | Electronics | |
| | Outdoor | |
| ics | Electronics | |
| | Unknown | |
| | Unknown | |
| ics | Electronics | |
| ics | Electronics | |
| ories | Accessories | |
| ics | Electronics | |
| ics | Electronics | |
| | Fashion | |
| | Outdoor | |
| ics | Electronics | |